

| Algorithm | Title                                                                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABC       | A powerful and efficient algorithm for numerical function optimization artificial bee colony (ABC) algorithm                                                               |
| ACROA     | ACROA Artificial Chemical Reaction Optimization Algorithm for global optimization                                                                                          |
| AHO       | The Archerfish Hunting Optimizer A Novel Metaheuristic Algorithm for Global Optimization                                                                                   |
| ALA       | Artificial lemming algorithm a novel bionic meta-heuristic technique for solving real-world engineering optimization problems                                              |
| AO        | Aquila Optimizer A novel meta-heuristic optimization algorithm                                                                                                             |
| ARO       | Artificial rabbits optimizationA new bio-inspired meta-heuristic algorithm                                                                                                 |
| BAEO      | Boomerang aerodynamic ellipse optimizer A human game-inspired optimization technique for numerical optimization and multilevel thresholding image segmentation             |
| BB-BC     | A new optimization method Big Bang–Big Crunch                                                                                                                              |
| BBO       | Beaver behavior optimizer A novel metaheuristic algorithm for solar PV parameter identification and engineering problems                                                   |
| BBO       | Biogeography-Based Optimization                                                                                                                                            |
| BCO       | An effective Bezier curve-based optimization (BCO) for large-scale numerical problems and 3D unmanned aerial vehicle path planning with efficient multiple threats evasion |
| BHO       | Bounty hunter optimizer A novel metaheuristic with an application to multi-UAV mobile edge computing and path planning                                                     |
| BMBO      | Solving01knapsackproblembyanovelbinarymonarch                                                                                                                              |
| BOA       | Billiards-inspired optimization algorithm a new meta-heuristic method                                                                                                      |
| BSO       | An integrated B-spline curves optimization framework with gradient learning capability for high-rise structural design and deep learning models                            |
| BWO       | Black Widow Optimization AlgorithmA novel meta-heuristic approach for                                                                                                      |
| CapSA     | A novel meta-heuristic search algorithm for solving optimization problems capuchin search algorithm                                                                        |
| CCO       | Centered Collision Optimizer A novel and efficient physics-based metaheuristic optimization algorithm for solving complex real-world engineering optimization problems     |
| CCO       | Cuckoo catfish optimizer a new meta-heuristic optimization algorithm                                                                                                       |
| CCS       | Convex combination search algorithm A novel metaheuristic optimization algorithm for solving global optimization and engineering design problems                           |
| CDO       | Chernobyl disaster optimizer (CDO) a novel meta-heuristic method for global optimization                                                                                   |
| CEO       | Cosmic Evolution Optimization A Novel Metaheuristic Algorithm for Numerical Optimization and Engineering Design                                                            |
| CFO       | A novel bio-inspired caterpillar fungus (Ophiocordyceps sinensis) optimizer for SOFC parameter identification via GRNN                                                     |
| CFO       | Cleaner fish optimization algorithm a new bio-inspired meta-heuristic optimization algorithm                                                                               |
| CJO       | Golden jackal optimization A novel nature-inspired optimizer for engineering applications                                                                                  |
| CMA-ES    | Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (CMA-ES)                                                             |
| COOT      | A new optimization method based on COOT bird natural life model                                                                                                            |
| CPA       | The colony predation algorithm                                                                                                                                             |
| CRO       | Chemical Reaction Optimization a tutorial                                                                                                                                  |
| CSA       | A novel metaheuristic method for solving constrained engineering optimization problems Crow search algorithm                                                               |

| Algorithm | Title                                                                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DA        | Dragonfly algorithm a new meta-heuristic optimization technique for solving single-objective, discrete, and multi-objective problems                             |
| DBA       | Detective Behavior Algorithm (DBA) A New Metaheuristic for Design and Engineering Optimization                                                                   |
| DE        | Self-adaptive differential evolution algorithm with discrete mutation                                                                                            |
| DMO       | Dwarf Mongoose Optimization Algorithm                                                                                                                            |
| DOA       | Dhole optimization algorithm a new metaheuristic algorithm for solving optimization problems                                                                     |
| DOA       | Dream Optimization Algorithm (DOA) A novel metaheuristic optimization algorithm inspired by human dreams and its applications to real-world engineering problems |
| DRA       | Divine Religions Algorithm a novel social-inspired metaheuristic algorithm for engineering and continuous optimization problems                                  |
| EAO       | Enzyme action optimizer a novel bio-inspired optimization algorithm                                                                                              |
| EMWPA     | Electromagnetic wave propagation algorithm A novel electromagnetic                                                                                               |
| EO        | Equilibrium optimizer A novel optimization algorithm                                                                                                             |
| EOSA      | Ebola Optimization Search Algorithm A New Nature-Inspired Metaheuristic Optimization Algorithm                                                                   |
| EPO       | Emperor penguin optimizer A comprehensive                                                                                                                        |
| FATA      | FATA An efficient optimization method based on geophysics                                                                                                        |
| FNO       | Far and Near Optimization A New Simple and Effective Metaphor-Less Optimization Algorithm for Solving Engineering Applications                                   |
| FOA       | A new Fruit Fly Optimization Algorithm Taking the financial distress model as an example                                                                         |
| FTO       | Fourier transform optimizer A novel physics-inspired metaheuristic algorithm for optimization problems                                                           |
| FTTA      | Football team training algorithm A novel sport-inspired meta-heuristic                                                                                           |
| FVIM      | Four vector intelligent metaheuristic for data optimization                                                                                                      |
| GBFIO     | A Novel Nature-Inspired Optimization Algorithm Grizzly Bear Fat Increase Optimizer                                                                               |
| GBO       | Gradient-based optimizer A new metaheuristic optimization                                                                                                        |
| GDA       | Golden Drop Algorithm A Water Wave and Golden Ratio inspired novel metaheuristic for global optimization.                                                        |
| GJA       | Gekko Japonicus Algorithm A Novel Nature-inspired Algorithm for                                                                                                  |
| GKSO      | Genghis Khan shark optimizer A novel nature-inspired algorithm for                                                                                               |
| GoldSA-II | A novel optimization method for solving constrained and unconstrained problems modified Golden Sine Algorithm.                                                   |
| GSA       | GSA A Gravitational Search Algorithm                                                                                                                             |
| GSK       | Gaining-sharing knowledge based algorithm for solving optimization problems a novel nature-inspired algorithm                                                    |
| GSO       | Glider snake optimizer (GSO) a nature-inspired metaheuristic algorithm for global and engineering optimization problems                                          |
| GTOA      | Group teaching optimization algorithm A novel metaheuristic method for solving global optimization problems                                                      |
| GWCA      | Great Wall Construction Algorithm A novel meta-heuristic algorithm for engineer problems                                                                         |
| GWO       | Grey Wolf Optimizer                                                                                                                                              |
| HBO       | Hannibal Barca optimizer the power of the pincer movement for global optimization and multilevel image thresholding                                              |
| HGS       | Hunger games search Visions, conception, implementation, deep analysis, perspectives, and towards performance shifts                                             |
| HGSO      | Henry gas solubility optimization A novel physics-based algorithm                                                                                                |

| Algorithm | Title                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HHO       | Harris hawks optimization Algorithm and applications                                                                                                                        |
| HLOA      | A novel metaheuristic inspired by horned lizard defense tactics                                                                                                             |
| HMS       | Human mental search a new population-based metaheuristic optimization algorithm                                                                                             |
| HS        | A New Heuristic Optimization Algorithm Harmony Search                                                                                                                       |
| HTSO      | Higher-Order Thinking Skills Optimizer A Metaheuristic Algorithm Inspired by Human Behavior and Its Application in Real-World Constrained Engineering Optimization Problems |
| HUS       | Human urbanization algorithm A novel metaheuristic approach                                                                                                                 |
| ICA       | Imperialist competitive algorithm An algorithm for optimization inspired by imperialistic competition                                                                       |
| JS        | A novel metaheuristic optimizer inspired by behavior of jellyfish in ocean                                                                                                  |
| KET       | A Kangaroo Escape Optimizer-Enabled Fractional-Order PID Controller for Enhancing Dynamic Stability in Multi-Area Power Systems                                             |
| KING      | KING An efficient optimization approach                                                                                                                                     |
| KLA       | Kirchhoff's law algorithm (KLA) a novel physics-inspired non-parametric metaheuristic algorithm for optimization problems                                                   |
| KOA       | Kepler optimization algorithm A new metaheuristic algorithm                                                                                                                 |
| LCA       | Liver Cancer Algorithm A novel bio-inspired optimizer                                                                                                                       |
| LEE       | LEE A physics-inspired optimizer based on LangEvin equation                                                                                                                 |
| LEO       | Language Education Optimization A New Human-Based Metaheuristic Algorithm for Solving Optimization Problems                                                                 |
| LGC       | Logistic-Gauss Circle optimizer Theory and applications                                                                                                                     |
| LiCOM     | Linear Constrained Optimization Method (LiCOM) A Novel Paradigm to Handle Engineering Optimization Problems                                                                 |
| LSO       | Leopard seal optimization (LSO) A natural inspired                                                                                                                          |
| MBO       | MBO Marriage in honey bees optimization a haplometrosis polygynous swarming approach                                                                                        |
| MFO       | Moth-flame optimization algorithm A novel nature-inspired heuristic paradigm                                                                                                |
| MShOA     | A Novel Bio-Inspired Optimization Algorithm Based on Mantis Shrimp Survival Tactics                                                                                         |
| MVO       | Multi-Verse Optimizer a nature-inspired algorithm for global optimization                                                                                                   |
| NDO       | Newton downhill optimizer with application to engineering optimization and breast cancer feature selection                                                                  |
| NOA       | Nizar optimization algorithm a novel metaheuristic algorithm for global optimization and engineering applications                                                           |
| NOA       | Nutcracker optimizer A novel nature-inspired metaheuristic algorithm for global optimization and engineering design problems                                                |
| NRBO      | Newton-Raphson-based optimizer A new population-based metaheuristic                                                                                                         |
| OA        | Orchard Algorithm (OA)A new meta-heuristic algorithm for solving                                                                                                            |
| OOA       | Octopus optimization algorithm a novel single- and multi-objective optimization algorithm for optimization problems                                                         |
| PBOA      | Perpendicular Bisector Optimization Algorithm (PBOA) A Novel Geometric-Mathematics-Inspired Metaheuristic Algorithm for Controller Parameter Optimization                   |
| PGA       | Phototropic growth algorithm A novel metaheuristic inspired from phototropic growth of plants                                                                               |
| PIMO      | Projection-Iterative-Methods-based Optimizer A novel metaheuristic algorithm for continuous optimization problems and feature selection                                     |
| PLO       | Polar lights optimizer Algorithm and applications in image segmentation and feature selection                                                                               |

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| PO        | Parrot optimizer Algorithm and applications to medical problems                                                                             |
| PO        | Puma optimizer (PO) a novel metaheuristic optimization algorithm and its application in machine learning                                    |
| PPO       | Escape after love Philoponella prominens optimizer and its application to 3D path planning                                                  |
| PSA       | PID-based search algorithm A novel metaheuristic algorithm based on PID algorithm                                                           |
| PSO       | Particle swarm optimization (PSO). A tutorial                                                                                               |
| PVS       | Passing vehicle search (PVS) A novel metaheuristic algorithm                                                                                |
| PWO       | Painted Wolf Optimization A Novel Nature-Inspired Metaheuristic Algorithm for Real-World Optimization Problems                              |
| RFO       | Rüppell's fox optimizer A novel meta-heuristic approach for solving global optimization problems                                            |
| RIME      | RIME A physics-based optimization                                                                                                           |
| SAA       | Snow avalanches algorithm (SAA) A new optimization algorithm for engineering applications                                                   |
| SequoiaOA | A sequoia-ecology-based metaheuristic optimisation algorithm for multi-constraint engineering design and UAV path planning                  |
| SFO       | The Sailfish Optimizer A novel nature-inspired metaheuristic algorithm for solving constrained engineering optimization problems            |
| SFOA      | Superb Fairy-wren Optimization Algorithm a novel metaheuristic algorithm for solving feature selection problems                             |
| SGA       | Snow Geese Algorithm A novel migration-inspired meta-heuristic algorithm for constrained engineering optimization problems                  |
| SIFO      | A memory-saving metaheuristic algorithm for onboard optimization Solitary Inchworm Foraging Optimizer                                       |
| SLC       | Soccer league competition algorithm A novel meta-heuristic algorithm for optimal design of water distribution networks                      |
| SMA       | Slime mould algorithmA new method for stochastic optimization                                                                               |
| SOA       | Seagull optimization algorithm Theory and its applications for                                                                              |
| SOA       | Skill Optimization Algorithm A New Human-Based Metaheuristic Technique                                                                      |
| SOO       | Stellar oscillation optimizer a nature-inspired metaheuristic optimization algorithm                                                        |
| SSO       | Salmon Salar Optimization A Novel Natural Inspired Metaheuristic Method for Deep-Sea Probe Design for Unconventional Subsea Oil Wells       |
| SWOA      | A small world algorithm for high-dimensional function optimization                                                                          |
| TGCOA     | A Novel Swarm Optimization Algorithm Based on Hive Construction by Tetragonula carbonaria Builder Bees                                      |
| TIEO      | Study on a new metaheuristic algorithm – Tribal intelligent evolution optimization and its application in optimal control of cooling plants |
| TJO       | Traffic jam optimizer A novel swarm-based metaheuristic algorithm for solving global optimization problems                                  |
| TLBO      | Teaching-Learning-Based Optimization An optimization method for continuous non-linear large scale problems                                  |
| TOA       | Tree optimization algorithm (TOA) a novel metaheuristic approach for solving mathematical test functions and engineering problems           |
| TOC       | Tornado optimizer with Coriolis force a novel bio-inspired meta-heuristic algorithm for solving engineering problems                        |
| TSA       | Tunicate Swarm AlgorithmA new bio-inspired based metaheuristic                                                                              |
| TUA       | Tactical unit algorithm A novel metaheuristic algorithm for optimal loading distribution of chillers in energy optimization                 |
| VPL       | Volleyball Premier League Algorithm                                                                                                         |

| Algorithm | Title                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| WCA       | Water cycle algorithm – A novel metaheuristic optimization method for solving constrained engineering optimization problems |
| WHO       | who-a-new-evolutionary-algorithm-bio-inspired-by-wildebeests-with-a-case-study-on-bank-customer-segmentation                |
| WHO       | Wild horse optimizer a new meta-heuristic algorithm for solving engineering optimization problems                           |
| WOA       | The Whale Optimization Algorithm                                                                                            |
| WOO       | Wave Optics Optimizer A novel meta-heuristic algorithm for engineering optimization                                         |
| WSO       | White Shark Optimizer A novel bio-inspired meta-heuristic algorithm                                                         |
| $\Phi$ PO | Philosophical proposition optimizer ( $\Phi$ PO) an epistemology-inspired algorithm for numerical optimization              |